

Integrated Intelligent Health Monitoring and Clinical Decision Support System

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AI Mastery Capstone — Mentor Defense

Healthcare | Machine Learning | Deep Learning | Generative AI | Agentic AI

1. Industry Problem and Motivation

Healthcare decisions can draw on both longitudinal clinical records and real-time wearable sensor data.

These information sources are often analyzed separately, limiting a unified decision-support workflow.

The project explores how multiple AI paradigms can be integrated without creating unsupported patient-level relationships.

Goal: provide transparent analytical support while preserving human clinical judgment and explicit safety boundaries.

2. Integration Rationale

How prior capstone work informed the final design

Statistical analysis

EDA, quantitative interpretation, limitations and bias awareness

Machine learning

Clinical readmission prediction and reproducible preprocessing

Deep learning

PyTorch LSTM sequence classification for wearable sensor windows

Generative AI

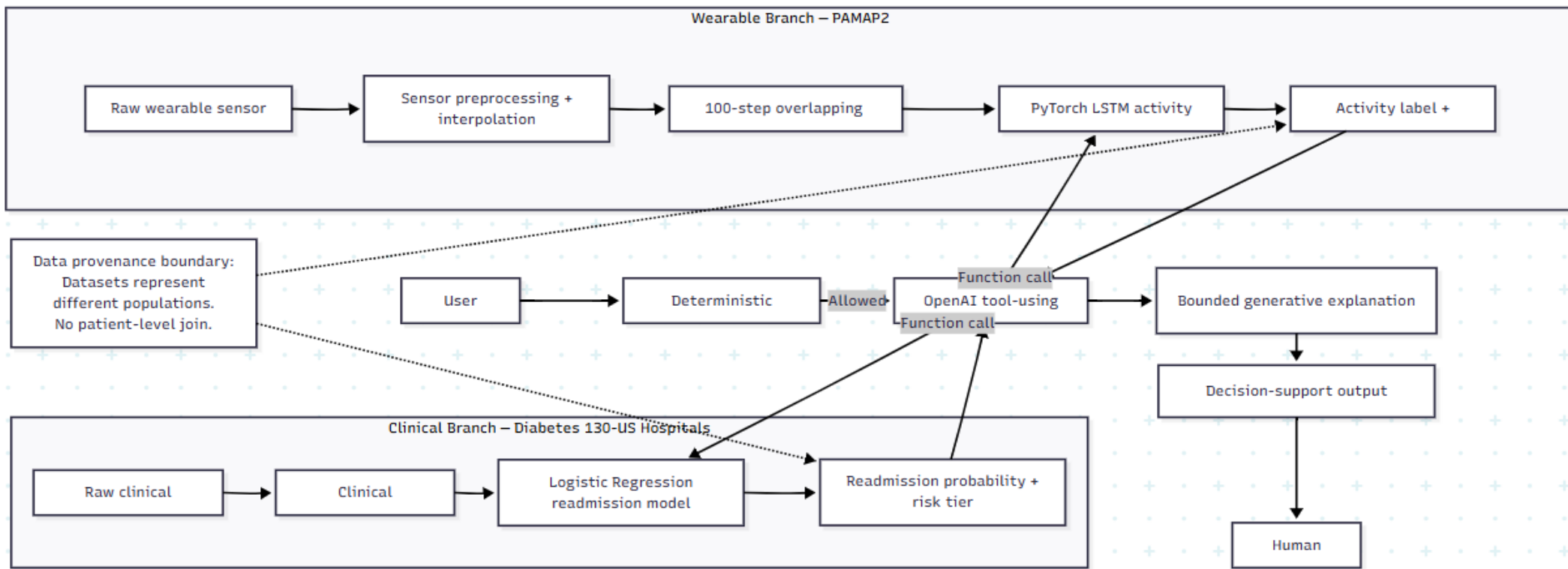
Grounded natural-language explanation of model-derived evidence

Agentic AI

OpenAI tool calling, orchestration, safeguards, and decision logging

3. Integrated System Architecture

Five major components: Clinical prediction pipeline, Wearable activity-recognition pipeline, OpenAI explanation layer, Agentic orchestration layer, Deterministic safety layer



4. Technical Decisions and Integration Choices

Clinical branch: Logistic Regression for a comparatively interpretable and computationally efficient readmission-risk baseline.

Wearable branch: LSTM over 100-step sensor windows to model temporal activity patterns.

The Diabetes and PAMAP2 datasets represent different populations; they are not merged into synthetic patient records.

Integration occurs at the application/orchestration layer, where model outputs remain independent analytical signals.

OpenAI function calling selects the appropriate local tool(s); the language model does not generate the numerical predictions.

5. Agentic Orchestration and Safeguards

Request validation occurs before agent execution.

Deterministic safety screening blocks unsupported diagnosis, prescribing, dosage, and treatment-selection requests.

The OpenAI agent can call `predict_readmission`, `classify_activity`, or both.

Local model outputs are returned as structured results with provenance.

The final explanation is bounded to decision support; decision logs make the execution path inspectable.

Responsibility Boundary

Models → predictions

OpenAI → orchestration + explanation

Safeguards → hard scope limits

Human professional → final judgment

6. Evaluation Results

Mixed evaluation: model performance + system behavior

Clinical ROC-AUC

0.669

Readmission model

Wearable Accuracy

0.571

Held-out subjects

Automated Tests

15 / 15

pytest suite

System Scenarios

5 / 5

Expected behavior

Clinical-only, wearable-only, and integrated routes executed the intended local tools.

Unsafe requests produced safety fallback without unnecessary model execution.

Invalid input was rejected before tool execution.

System-level success measures orchestration behavior; it does not imply perfect predictive accuracy.

7. Failure Cases, Limitations, and Ethical Risks

Observed wearable failure: an unseen-subject example was misclassified (lying → ironing), showing that correct orchestration does not guarantee a correct model prediction.

PAMAP2 contains only nine participants, limiting generalizability.

The same held-out subject group was used during intermediate validation and final evaluation; it is not a completely untouched final test set.

Clinical and wearable datasets represent different populations and cannot support patient-level causal conclusions.

OpenAI explanations are non-deterministic; the prototype also lacks production privacy, security, monitoring, clinical validation, and regulatory governance.

8. Professional Relevance

Demonstrates system-level thinking beyond training a single model.

Combines data preprocessing, supervised ML, deep learning, generative AI, and agentic orchestration.

Uses modular code, persisted models, tool interfaces, automated tests, provenance, and decision logging.

Shows responsible technical judgment: unsupported dataset merging was avoided and medical capability boundaries were enforced.

Reflects real-world AI responsibilities: performance, reliability, transparency, maintainability, safety, and governance.

9. Next Steps and Defense Summary

Use larger and more representative clinical and wearable populations.

Adopt separate participant-level train, validation, and final-test evaluation or participant-level cross-validation.

Add fairness analysis, drift monitoring, authentication, access controls, audit processes, and privacy/compliance review.

Perform prospective clinical validation before any real-world decision-support use.

Core design principle: keep prediction grounded in specialized models, orchestration transparent, and final responsibility with humans.

Capstone contribution: a tested, bounded, multi-model healthcare AI prototype—not an autonomous clinical system.